

PREFABRICATED CABIN SPECIFICATIONS

Scope of Work: Fabrication, Supply and Install of Pre-Fabricated Cabins for Retail Outlets. Cabins shall be fabricated at vendor's factory and assembled at Retail Outlets. Welding or hot work shall not be allowed at Retail outlets, all pre work shall be completed at vendor's factory.

GENERAL DESIGN AND CONSTRUCTION:

1. SHELL:

The shell structure shall comprise of sidewalls, end walls (All four sides) and self-draining roof.

2. BASE FRAME / FLOOR SYSTEM:

The Prefab Cabin shall be mounted on a base frame of C Channel (150x75x8mm thick). Base frame will be fixed on the platform provided by BPCL.

3. SIDE & END WALLS:

The sidewalls shall be fabricated out of 1.6 mm (Minimum) thick steel sheet duly punched and welded in uniform panels. The four corner posts of 100 x 100 x 50 by 3.2 mm sheet shall be fixed on the base frame and sufficiently strengthened for stability. All the panels shall be POWDER COATED (electrostatic powder coated of 25 micron thick & in approved shade & colour), fully glazed.

All materials conforming to IS: 1079 Grade O/ IS: 2062 Grade A.

4. ROOF

Self-draining type Trapezoidal roofing profile made out of 0.55mm TCT with high tensile bare Galvalum sheet of 275 mpa min yield strength of grade AZ-150 (55% Aluminum, 43.5% Zinc and 1.5% Silicon) with aluzinc coating of 150 GSM. The roofing should be Clip-on Type. , these sheets shall be fixed on C / Z purlins mounted on the Panels with extra strengthening as required. Roof shall be adequately cambered for draining of rain water.

5. Outside Fascia Branding:

Three Pillars and Top fascia on two sides to be made out of pre-coated sheet of 0.80mm TCT with high tensile bare Galvalum sheet of 540 mpa min yield strength of grade AZ-150 (55% Aluminum, 43.5% Zinc and 1.5% Silicon) with aluzinc coating of 150 GSM. The pre-coated sheet should be finished with 18-22 micron polyester coating (on the exposed surface) with BPCL yellow colour over epoxy primer of minimum 5 micron. The size of box shall 300x275x25 mm as per layout shown in drawing. Fascia shall be fixed with self-tapping counter sunk screws.

6. SURFACE TREATMENT AND PAINTING:

The surface preparation of all fabricated steel components shall be done by manual grit blasting to achieve SA-2 quality finish, followed by one coat of zinc based primer and two coats of epoxy paint by air spray method. The under frame shall be painted with Bituminous / red lead paint of reputed make.

7. INSULATION:

Average 40 mm thick thermal insulation with resin bonded fiber glass wool having density of 16 kg/m3 and conforming to IS: 8183 shall be provided on the side walls, end walls (All four side).

8. INNER PANELLING:

Inside paneling shall be of 8 mm cement Fiber Board or HDF Board, fixed on the steel panels with self-tapping screws. The internal shall be aesthetically finished so to give custom built appearance with high quality workmanship. All vertical and horizontal corners shall be neatly and smooth finish. 2 Coats of Primer and 3 Coats of Plastic Paints will be applied on the inner Wall Panels.

Suitable provision shall be made to provide Split AC, indoor and outdoor units on rear wall of cabin for future. Supply of AC is not in the scope of this tender.

9. DOOR & WINDOWS:

All aluminum works shall be POWDER COATED (electrostatic powder coated of 25 micron thick & in approved shade & colour), fully glazed, using heavy duty aluminum extruded section, glazed with gasket, felt & aluminum beading, including decorative door handles per leaf both inside and outside, hydraulic door closure (heavy duty & approved make) for door closure, locking arrangement (of Godrej make), tower bolts, other fixtures & fastenings for door & windows, sealing clips, rubber lining for air tightness, etc complete in all respect.

Frame: Aluminum section size 45mm x 45mmx1.8mm (4577) with PVC beading & glazing clip of 0.071 for doors and windows. The door Frame should be 63.5 x 45 x 1.8mm.

Door : Heavy duty door frame of vertical, top and bottom stile with 5mm thick glass.

Sliding Window: Sliding window with heavy duty frame and sections, two track sections with 5 mm thick clear float glass.

Fixed window: Fully glazed windows with 6 mm Toughened Glass with reflective coating for minimum 40% dissipation of heat shall be used for Fixed Windows, Door and Sliding Windows.

10. Flooring :

Bottom frame shall be made of 48 x 48 x 2 mm thick MS Square tubes and fixed to base frame using Nuts and Bolts making at a maximum distance of 600 mm. 1 mm GI Sheet shall be fixed by screws on these Square Tubes to ease laying of PCC/ Tile Fixing Adhesive. Vitrified flooring tiles of size 600mmx600mmx8mm (Kazaria, Johnson or Equivalent), in wooden finish, shall be fixed on the PCC/ Tile Fixing Adhesive.

11. False Roofing :

False ceiling shall be done in 8 mm PVC coated and insulated 600x600 mm gypsum tiles of reputed make. False ceiling to be provided at height of 9ft with suitable suspending arrangement.

12. ELECTRICALS:

PVC insulated copper wire of ISI mark shall be used for the wiring. This wiring shall be done in PVC/ flexible conduits. As a measure of safety, continuous copper earth wire shall be provided as necessary. For lighting circuits, 1.5 sq mm cable and for heavy-duty circuits, such as air conditioners, heating circuits, 4 sq mm cable shall be used. DOL switches of capacity 250V/15 Amps shall be provided for air conditioners. For connecting power supply to bunk houses, 3 pin input industrial socket of 32 Amps capacity shall be provided.

All the lamps, fans, exhaust fans, plug points etc. shall have independent slash/piano type switches of required rating inside the cabin. Ceiling fans/False ceiling light shall be of Crompton, or equivalent of reputed make. All electrical points & switches shall be of Havels/ Anchor or equivalent make. Necessary circuit breakers i.e. MCB & ELCB shall be provided outside cabin to connect the power supply.

Supply to Air conditioner shall be through metal clad MCB of adequate rating. Junction Box shall be provided outside on the prefab house for telephone connections with suitable connecting points inside the prefab cabin.

List of approved makes

- Steel plates: Jindal, Bhushan ,TATA,SAIL
- Base frame steel: Jindal, Bhushan ,TATA,SAIL
- Cement Fiber board: Everest, RAMCO or equivalent
- Aluminum Sections: Hindalco, Jindal, Indal
- Glass Make: AKAI, MODIGUARDA, Saint Gobain or equivalent
- Flooring tiles: Kazaria, Johnson or Equivalent
- Galvalume sheet: Bhushan, Jindal or TATA Blue scope
- Copper cables: Polycab , Finolex or equivalent Make
- MCB: Havelts, LT or equivalent
- LED lights: Philips, Bajaj, Crompton or equivalent
- Ceiling Fan: Bajaj, Usha, Crompton or equivalent
- Switches: Philips, Havells or equivalent
- Plywood and Laminates: Formica, Sunmica, Bravia etc
- Chair rolling type- supreme/Godrej or equivalent.